

HARSHITA KEERTHIPATI

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EDUCATION

University of Washington

Seattle, Washington

BS in Computer Science (GPA: 3.7)

June 2028

Courses: Multivariable Calculus, Introduction to Object Oriented Programming, Data Structures & Algorithms, Discrete Mathematics/Logic, Linear Algebra, Data Science, Software Design Implementation, Hardware/Software Interface

Orgs: Lavin Entrepreneurship Program, AWS ReachBack Cohort, Beliveau Lab, CSEED Buildspace, SWE, WeLead, DubHacks 2025

EXPERIENCE

Salesforce

San Francisco, CA

Software Engineer Intern

Summer 2027

- Incoming Software Engineer Intern on Salesforce's Data 360 organization.

Shopify

Bellevue, WA

Intern Engineer

Winter 2027

- Incoming Intern Engineer at Shopify.

Dharma

Pleasanton, CA

Co-Founder

May 2025 - Present

- Developing a full-stack SwiftUI iOS application for AI-powered spiritual guidance & personalized user experiences.
- Designing RAG pipelines with Supabase, PostgreSQL, & Redis to retrieve relevant content, maintain real-time user state, & deliver low-latency personalized AI responses with a median latency of 2.3 seconds.
- Building & testing FastAPI backend services & LLM-powered features using OpenAI's API, including personalized verse recommendations & context-aware chat workflows using a database of 5,000+ verses.

UW Husky Satellite Lab

Seattle, WA

Software Engineer

October 2025 - Present

- Implementing C++ star-identification algorithms in the LOST star-tracker pipeline, extracting centroids from noisy CCD images & matching observed star fields to onboard catalogs using geometric pattern matching & efficient catalog indexing.
- Developing constellation-matching algorithms based on inter-star angular distance invariants, implementing tetrahedral-based pattern matching with hash-table & range-query indexing for lost-in-space identification.
- Building & evaluating ML-assisted algorithms to filter & rank candidate star-catalog matches using gradient-boosted models trained on angular separation residuals, magnitude consistency, centroid uncertainty, & false star detections.

Beliveau Lab @ UW Medicine

Seattle, WA

Undergraduate Researcher

January 2026 - Present

- Training & evaluating a 3D U-Net in PyTorch to learn a volumetric image-to-image mapping from DAPI microscopy images to red & green FUCCI fluorescence channels to enable segmentation-free analysis of cell-cycle patterns.
- Designing a differentiable, frequency-aware loss function that combines pixel-wise MSE with radially weighted Fourier-domain loss to better capture biologically meaningful spatial patterns & retain fine image details.
- Developing reproducible GPU-based ML experimentation pipelines on a Sun Grid Engine cluster, including automated hyperparameter sweeps, early stopping, checkpointing, TensorBoard monitoring, & preprocessing using bead detection, Gaussian fitting, KD-tree matching, & polynomial regression.

California Business Technology (CBT)

Pleasanton, CA

Information Security Engineer

June 2024 - May 2025

- Selected as California's first Cybersecurity Youth Apprentice, completing 31 state-defined cybersecurity competencies across vulnerability assessment, secure systems architecture, & threat mitigation.
- Conducted penetration testing/vulnerability assessments using Greenbone/OpenVAS, Nmap, & Wireshark to identify misconfigurations, exposed services, & network-level weaknesses; deployed Docker containers & conducted TLS/SSL inspection to evaluate encryption & certificate security.
- Built secure integrations with HubSpot CRM using Python & React to manage sensitive company data; simulated defenses against ransomware payloads, DDoS, & botnet intrusion attempts to assess system resilience.

PROJECTS

GPT Transformer: Built a GPT-style transformer from scratch in PyTorch using Jupyter, implementing tokenization, masked multi-head self-attention, positional encodings, layer normalization, & full autoregressive training loops. Trained the model end-to-end to study how transformer architectures learn, converge, & generate coherent text at scale.

Fitcheck: Developed virtual try-on iOS application using FLUX-based diffusion models in PyTorch deployed on AWS (ECS, EC2, S3, SQS). Built LLM-based garment classifiers (90% accuracy) & conducted 50+ user interviews to guide retraining.

Nenu Apparel (Buildspace S5): Built & operated a direct-to-consumer clothing brand on Shopify, integrating Shopify Payments, Google Analytics, & email marketing automation. Developed & optimized the e-commerce experience using React, JavaScript, Bootstrap, & HTML/CSS, with six consecutive weekly product drops & 50+ pre-orders in a single launch cycle.

Findar (WINFO Hackathon 2026): Developed AR-based passive object tracking system using LiDAR, YOLO object detection, & 360° camera feeds to build a real-time 3D model of indoor environments with continuous location history. Developed a native Swift iOS app to query & retrieve last-seen object positions without manual tagging.

SKILLS

Languages: Python, Java, C++, Swift, JavaScript, SQL, R, HTML/CSS

Frameworks & Libraries: React.js, Node.js, FastAPI, Django, Flask, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

Tools & Platforms: AWS, Docker, Linux, Git, Xcode, PostgreSQL, MySQL, Supabase, OpenAI API, Tailwind CSS